The $M(e^+e^-)$ lineshape of $J/\psi \to e^+e^-(\gamma)$: a corrected fit function and how to use it

J/ψ electroproduction RC session, for V. Kubarovsky

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Abstract

The function `dNdMee.cpp` describes the e^+e^- invariant-mass spectrum of $J/\psi \to e^+e^-(\gamma)$ with three parameters: the peak position M, a Gaussian resolution σ, and the fraction `br` of decays with a photon above $\varepsilon = 1$ MeV. Fitted to the generator's own J/ψ decays it finds M exactly, but its tail is off by up to 20 % and, folded with a 30 MeV resolution, it returns a J/ψ yield 2.5 % too low and M 0.7 MeV too high with a perfectly acceptable χ^2 — an error a fit to data cannot reveal. Three causes are identified: the photon energy is taken as $M - M_{ee}$ instead of $k = (M^2 - M_{ee}^2)/2M$, the spectrum is first order instead of exponentiated, and its normalisation constant is 3 % off. A corrected function `dNdMee_v2.cpp`, with the same interface, agrees with the generator to five digits, returns M, σ and the yield without bias, and does not depend on ε. The recommendation for data is to keep the radiative part fixed by QED and fit only the yield, the mass scale and the resolution; the detector response beyond a Gaussian is the next step, with GEMC.

1 The spectrum

In the J/ψ rest frame a photon of energy k leaves the pair with

$$M_{ee}^2 = M^2 - 2Mk, \qquad k = \frac{M^2 - M_{ee}^2}{2M}, \qquad \frac{dk}{dM_{ee}} = \frac{M_{ee}}{M}. \tag{1}$$

To first order in α the photon spectrum is (Ehlotzky and Mitter, $E = M/2$, $x = k/E$)

$$\frac{1}{\Gamma_0}\frac{d\Gamma}{dk} = \frac{2\alpha}{\pi}\Big[2\ln\Big(\frac{2E}{m_e}\sqrt{1-x}\Big) - 1\Big]\frac{1}{k}\Big(1 - x + \frac{x^2}{2}\Big), \tag{2}$$

and the correction with all photons below K (soft and virtual included) has the closed form $\delta(K)$ of `rc_vector_mesons.py:delta_vm`, with $d\delta/dK$ equal to Eq. (2). The leading logarithm, $\delta(K) \simeq \beta\ln(K/E)$ with

$$\beta = \frac{2\alpha}{\pi}\Big(\ln\frac{M^2}{m_e^2} - 1\Big) = 0.0763, \tag{3}$$

is exponentiated, giving the cumulative distribution of the photon energy

$$F(K) = \Big(\frac{K}{E}\Big)^{\beta}\,\frac{1 + \delta(K) - \beta\ln(K/E)}{1 + \delta(E)}, \qquad F(E) = 1. \tag{4}$$

F is normalised to the radiation-inclusive width: the measured $\mathcal{B}(J/\psi \to e^+e^-)$ counts $e^+e^-(\gamma)$ (BESIII restores the tail below its 2.7 GeV window with PHOTOS), so the lineshape is a probability density and carries no factor $1 + \delta$. This is exactly what the event generator samples (`exclurad_py/generator/decay_rad.py`); its angular part, the exact tree-level matrix element, does not affect M_{ee}. Two useful numbers: $1 - F(1\text{ MeV}) = 0.3945$ of the decays have a photon above 1 MeV, and $P(M_{ee} < 3.0\text{ GeV}) = 0.1466$.

2 What is wrong with `dNdMee.cpp`

The original function uses Eq. (2) with k replaced by $x = M - M_{ee}$, a Gaussian peak of weight $1 - \texttt{br}$, and the tail scaled by $\texttt{br} \times \texttt{Norm}$, $\texttt{Norm} = 2.0707$. Three points:

1. **Wrong variable.** $M - M_{ee}$ is not the photon energy; with Eq. (1), $k = (M - M_{ee})(M + M_{ee})/2M$, smaller by up to 20 % at $M_{ee} \simeq 2$ GeV, and a density in M_{ee} needs the Jacobian M_{ee}/M.

2. **First order.** The $1/k$ spectrum has too many photons just above ε; the exponentiated $k^{\beta-1}$ shape matters because $\varepsilon = 1$ MeV is deep in the soft region.

3. **Normalisation.** $\int_{1\,\mathrm{MeV}}^{E}$ of Eq. (2) is 0.4984, so the normalising constant should be 2.0066, not 2.0707: the tail is normalised to 1.032.

Fig. 1 compares the two unfolded tails at the same `br`: the original is 5–10 % low over most of the range and up to 35 % high next to the peak.

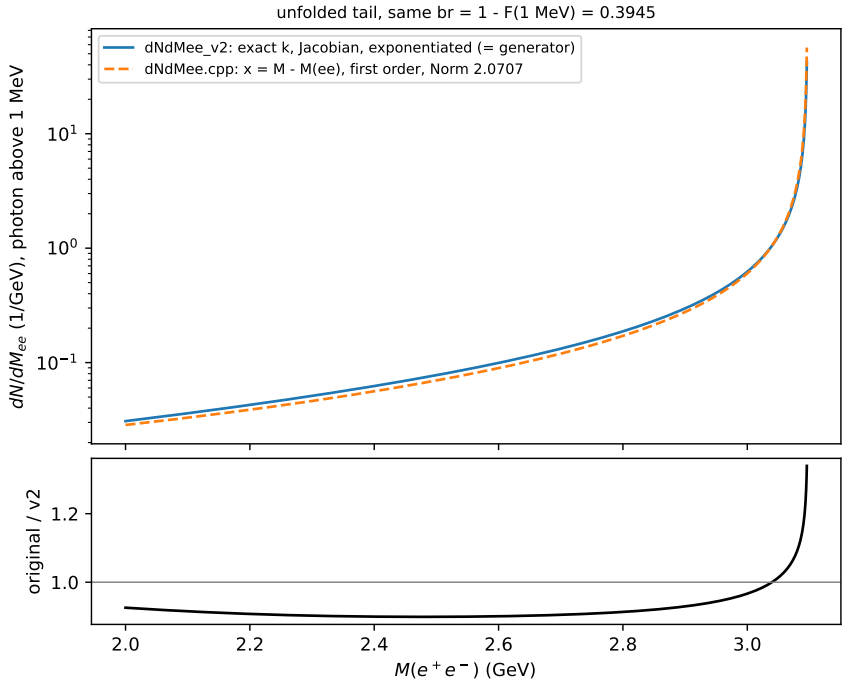


Figure 1: Unfolded M_{ee} tail (photon above 1 MeV) of the corrected function (= generator) and of the original formula, both with $\texttt{br} = 1 - F(1\ \mathrm{MeV})$.

3 The corrected function `dNdMee_v2.cpp`

Same call, `dNdMee(mee, M, sigma, br)`, plus a general `dNdMee_full(mee, M, sigma, br, eps, m)` (lepton mass m, so it also serves $\mu^+\mu^-$). It returns dN/dM_{ee} in GeV^{-1}, normalised to one:

$$\frac{dN}{dM_{ee}} = F(\varepsilon)\, G(M_{ee} - M; \sigma) + \frac{\texttt{br}}{1 - F(\varepsilon)} \int_{\varepsilon}^{k_{\max}} \frac{dF}{dk}\, G\big(M_{ee} - \sqrt{M^2 - 2Mk}; \sigma\big)\, dk, \tag{5}$$

with dF/dk from Eq. (4) in closed form, the integral done in $\ln k$ (Simpson, 400 nodes, within $\pm 8\sigma$). **`br` < 0, the default, means the QED value** $1 - F(\varepsilon)$; a value in $[0, 1]$ overrides it and is meant for

studies only. $\sigma \leq 0$ returns the unfolded tail. The code has no dependence outside the C++ standard library and compiles also as a shared library (used for every fit below, so the fits test the code a user will call).

Checks: $F(1\text{ MeV}) = 0.60551$, and $P(k > K)$ for $K = 0.01$, 0.1, 0.3, 1 GeV is 0.27855, 0.14368, 0.07724, 0.01590, identical to five digits to the generator; $\int dN/dM_{ee} = 0.9965$ over 1.0–3.4 GeV at $\sigma = 30$ MeV (the rest is below 1 GeV).

4 Tests

4.1 MC truth, no resolution

The generator's M_{ee} (200k Born decays, no acceptance cut) fitted with $\sigma = 0.1$ MeV fixed and N, M, `br` free (Fig. 2, 0.2 MeV bins at the peak; χ^2 excludes the 1.2 MeV next to the peak, which a 0.1 MeV Gaussian cannot describe for a delta):

model	M (GeV)	`br`	χ^2/ndf
original `dNdMee.cpp`	3.096900	0.3802 ± 0.0011	1302/215
exact k + exponentiated (= v2)	3.096900	0.3944 ± 0.0011	202/215
counted in the sample	3.0969	0.3946	

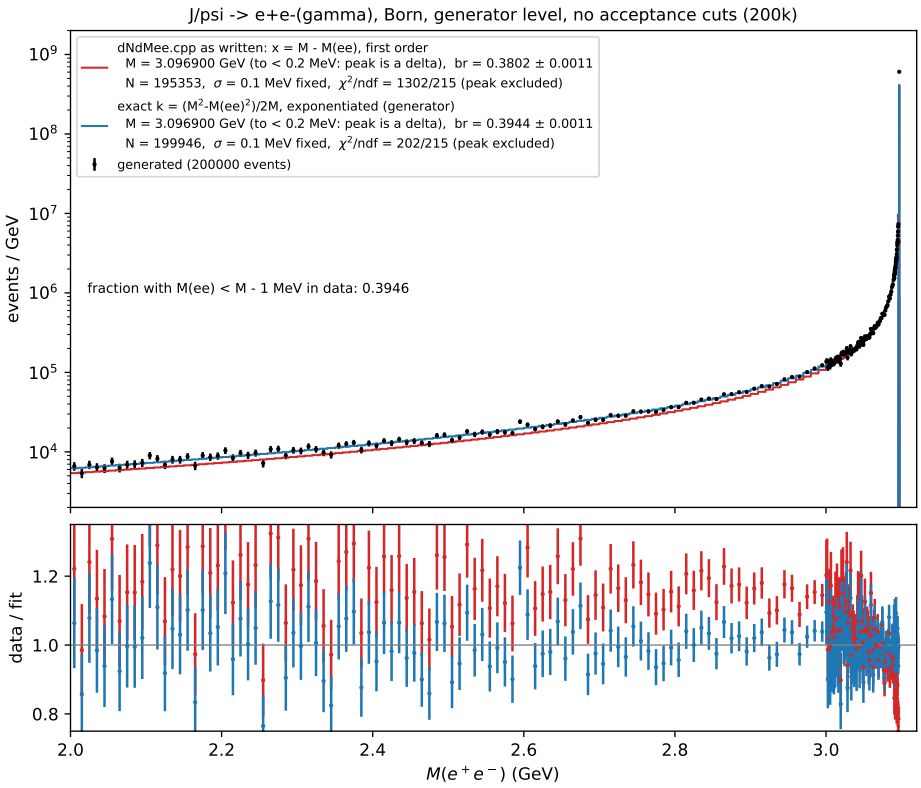


Figure 2: Generator-level M_{ee} with the original (red) and the exact (blue) model; lower panel data/fit.

The same MC truth, without any resolution and with no free parameter in v2 (N = number of generated decays, `br` = QED), is shown in Fig. 3 against $\Delta M = M - M_{ee}$ in log bins. The generator's threshold $\varepsilon = 1$ MeV is directly visible: 121 083 decays (0.6054) sit exactly at M, against $F(1\text{ MeV}) =$

0.6055 (121 102 expected); not a single event lies between $\Delta M = 0$ and $\Delta M(k{=}\varepsilon) = 1.0002$ MeV; above it v2 describes the tail over three decades of ΔM with $\chi^2/\mathrm{ndf} = 50/61$. The original function, even at its own best fit, is 20 % low at $\Delta M = 1$–3 MeV and 15 % high above 0.3 GeV — the $1/k$ first-order spectrum with k replaced by $M - M_{ee}$.

4.2 Pseudo-data: MC truth folded with a Gaussian

No GEMC and no data are used in this section. The input is the generator's own M_{ee} (the 200k events of the previous section); each event is shifted by a random number drawn from a Gaussian of width $\sigma_{\mathrm{true}} = 30$ or 50 MeV. This is an idealised detector with a known answer: $M = 3.0969$ GeV, $\sigma = \sigma_{\mathrm{true}}$, $N = 200\,000$ (the function is normalised over all M_{ee}, so N is the total number of decays even though the fit range is 2.6–3.3 GeV).

σ_{true}	fit	M (GeV)	σ (MeV)	N	`br`
30 MeV	F1: v2, br = QED	3.09685	29.92 ± 0.07	$199\,890 \pm 460$	0.3945 fixed
	F2: v2, br free	3.09692	29.89 ± 0.07	$199\,980 \pm 470$	0.398 ± 0.003
	F3: original, br free	**3.09762**	29.73	**195 035**	**0.438**
50 MeV	F1: v2, br = QED	3.09679	49.91 ± 0.11	$199\,940 \pm 460$	0.3945 fixed
	F2: v2, br free	3.09700	49.83 ± 0.12	$200\,100 \pm 470$	0.400 ± 0.003
	F3: original, br free	**3.09790**	49.64	**195 120**	**0.446**

χ^2/ndf is 125/120 (F1), 123/119 (F2) and 137/118 (F3) at 30 MeV and 125/134, 121/133, 122/132 at 50 MeV: **the original function fits as well as the correct one**, while it loses 2.5 % of the yield and shifts M by +0.7–1.0 MeV. The corrected function returns all three quantities without bias, and a free `br` comes out at the QED value.

4.3 Dependence on ε

With `br` fixed to $1 - F(\varepsilon)$ at every ε (Fig. 6): σ stays at 30.02–30.05 MeV and χ^2 at 116–117/120 for $\varepsilon = 0.1 \ldots 10$ MeV, while `br` itself moves from 0.49 to 0.28 — it is a bookkeeping split, not a physics parameter. M shifts by -0.25 MeV at $\varepsilon = 5$ MeV and -0.55 MeV at 10 MeV, because decays with a photon below ε are put exactly at M; it is stable within 0.06 MeV for $\varepsilon \leq 1$ MeV.

4.4 Lepton acceptance

The 1M Born sample of the f18in production carries the generator's lepton cuts (`iacc jpsi`). Folded with 30 MeV: F1 gives $M = 3.09697$ GeV, $\sigma = 29.95 \pm 0.03$ MeV, χ^2/ndf 117/122; a free `br` gives 0.3949 ± 0.0011, the QED value. The cuts remove radiative decays slightly more often (13.8 % below 3.0 GeV instead of 14.7 % in that sample) but do not change the shape a fit sees (Fig. 7).

5 The branching ratio, Γ_{Born} and vacuum polarisation

What the PDG value is. $\mathcal{B}(J/\psi \to e^+e^-) = 5.971\,\%$ is radiation-inclusive: it counts every decay to e^+e^- with any number of photons. The separate entry $\mathcal{B}(e^+e^-\gamma,\ E_\gamma > 100\ \mathrm{MeV}) = (8.8 \pm 1.4) \times 10^{-3}$ is a subset of it, not an additional channel. The ratio of the two, 0.147 ± 0.023, agrees with the QED fraction $1 - F(0.1\ \mathrm{GeV}) = 0.1437$ of Eq. (4), which predicts $0.1437 \times 5.971\,\% = 8.58 \times 10^{-3}$. The inclusive width is the right quantity for a table whose branching ratios must sum to one, it does not depend on a detector window, and it is free of the electron-mass logarithms of final-state radiation (KLN theorem).

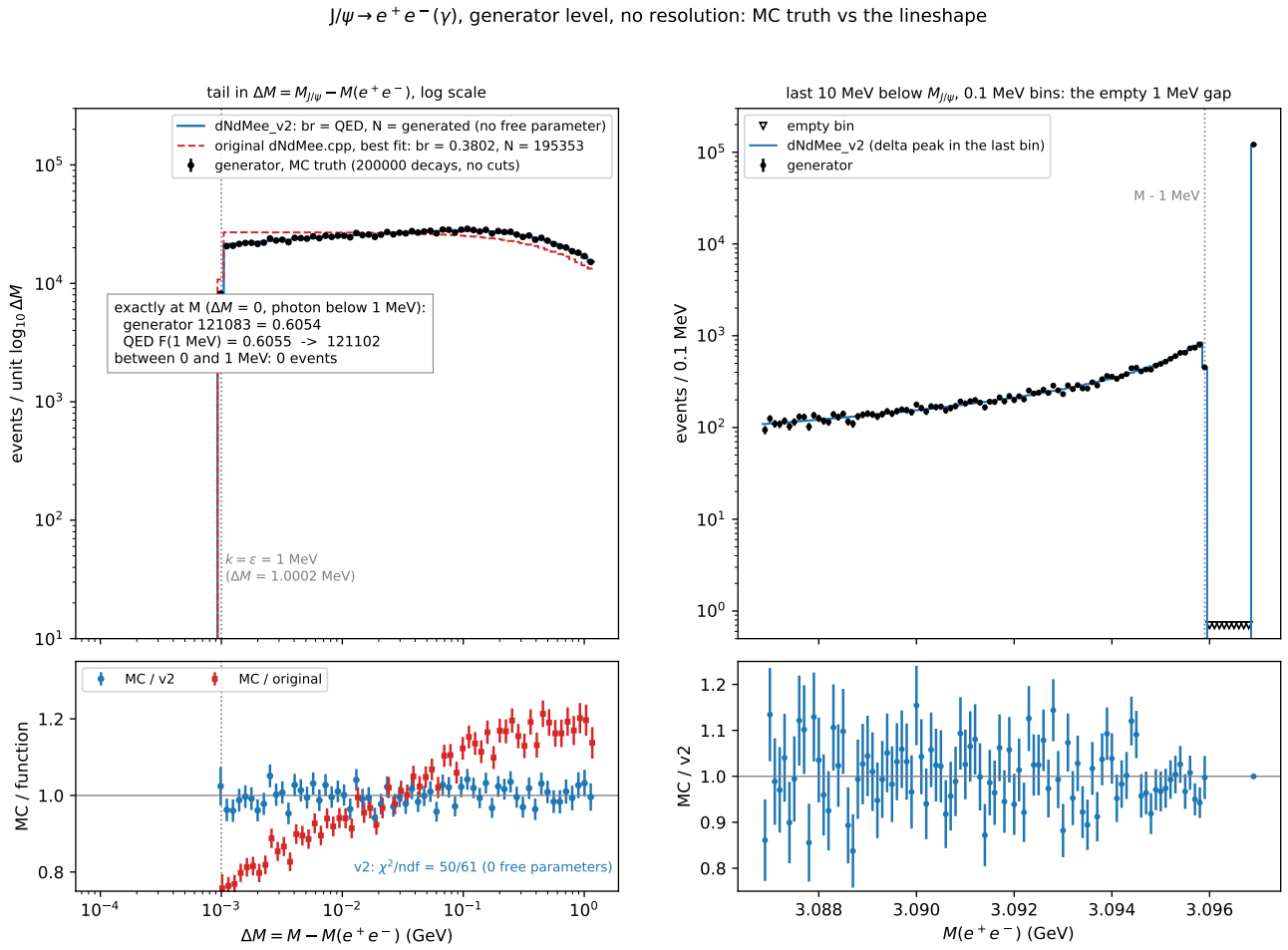


Figure 3: MC truth without resolution. Left: tail in $\Delta M = M - M_{ee}$ (log bins), v2 with no free parameter and the original function at its best fit. Right: the last 10 MeV below M in 0.1 MeV bins: the empty 1 MeV gap and the delta peak.

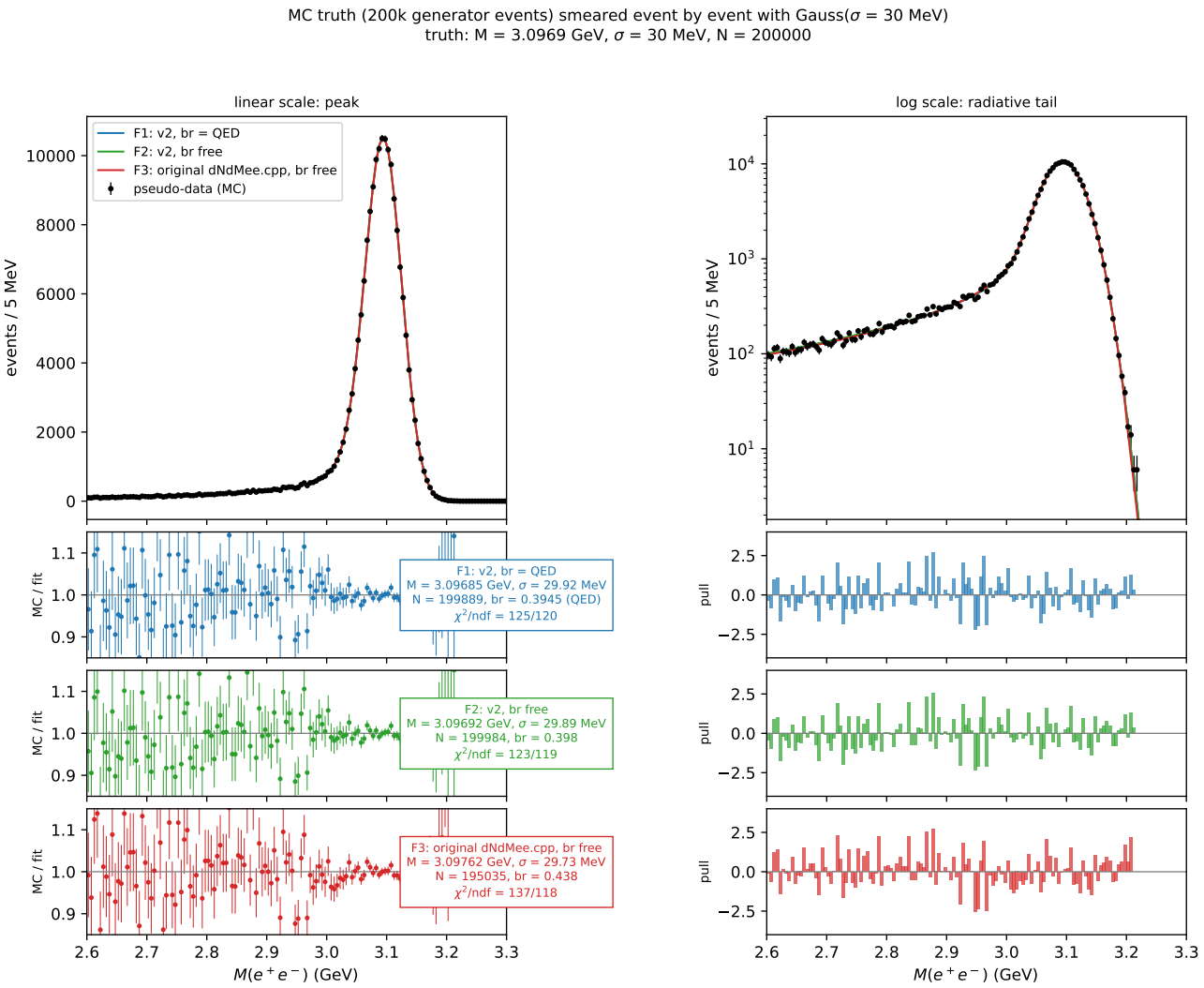


Figure 4: How the three fits lie on the MC: pseudo-data ($\sigma_{\text{true}} = 30$ MeV) in linear and log scale, and for each fit MC/fit and the pull per bin. All three describe the MC within statistics; F3 does it with the wrong M and N.

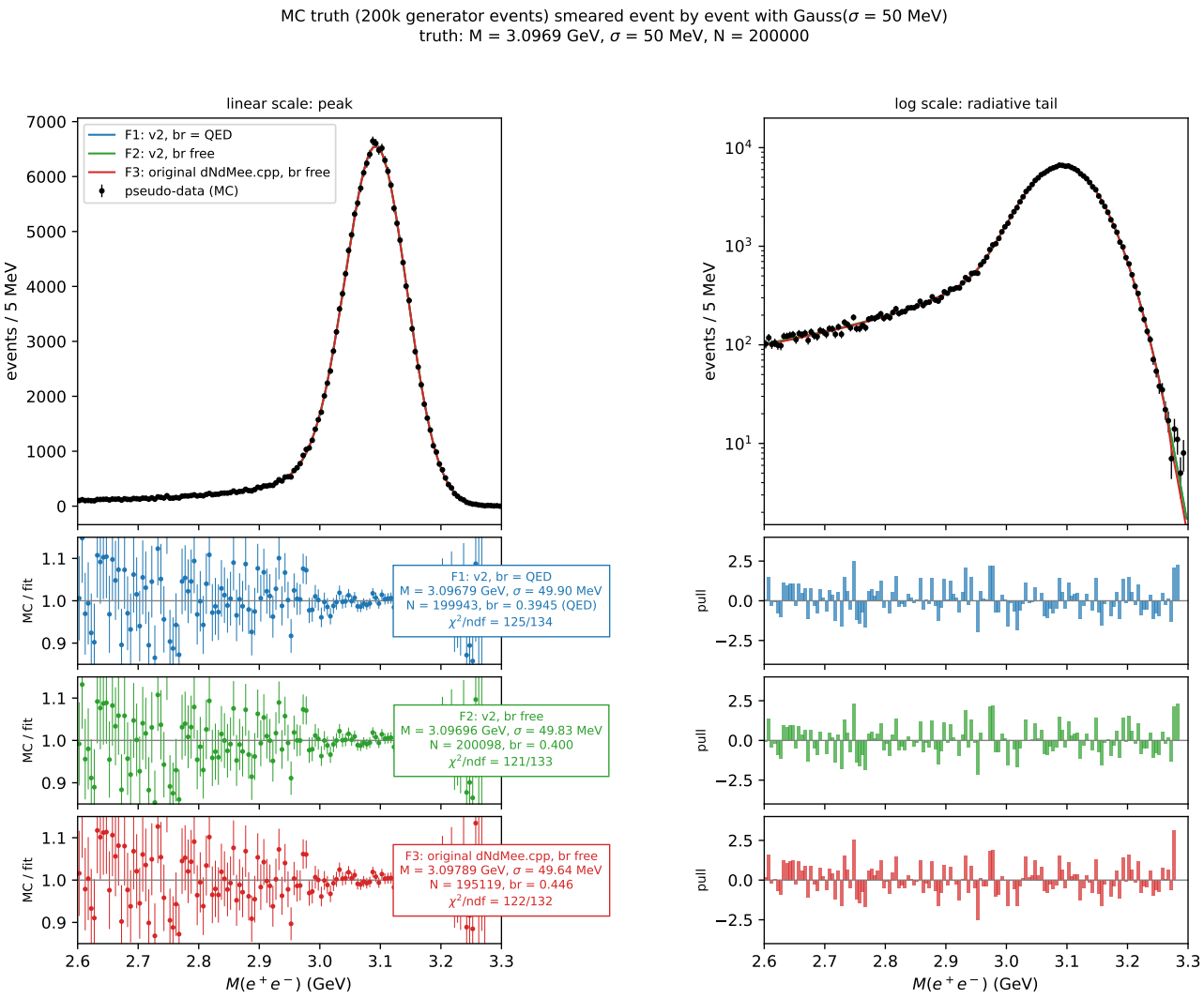


Figure 5: As Fig. 4, $\sigma_{\text{true}} = 50$ MeV.

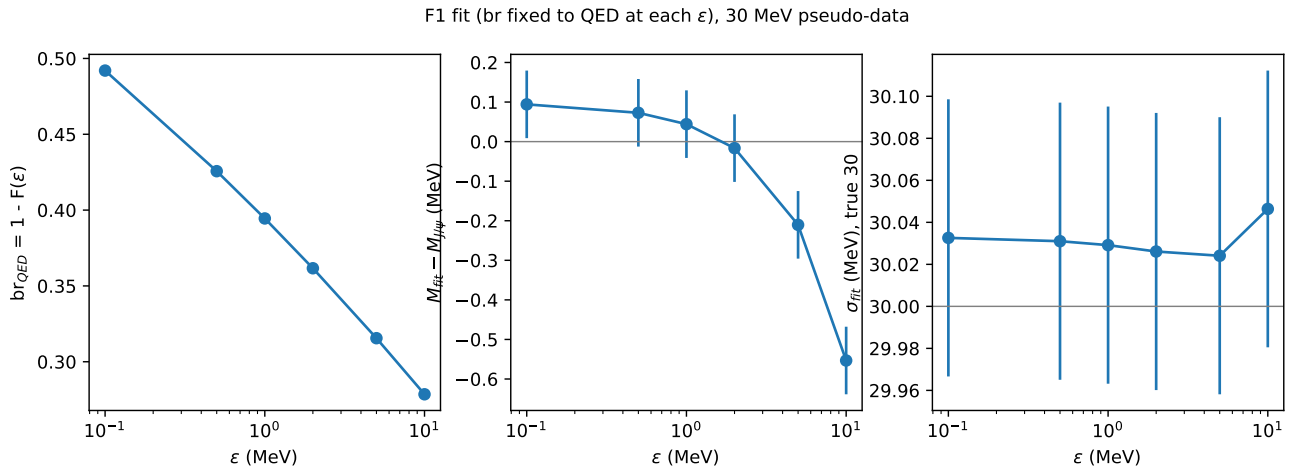


Figure 6: QED `br`, fitted M and fitted σ against ε.

$\Gamma_{\rm Born}$ exists and can be computed. The lowest-order width — the quantity of potential models, $\Gamma_0 \propto \alpha^2 Q_c^2 |\psi(0)|^2/M^2$, and of lattice decay constants, $\Gamma_0 = 4\pi\alpha^2 Q_c^2 f_\psi^2/3M$ — follows from the measured one once *all* lepton-side QED corrections, loops included, are divided out:

$$\Gamma = \Gamma_{\rm Born}\left(1+\frac{3\alpha}{4\pi}\right)\frac{1}{|1-\Pi(M^2)|^2}, \qquad \frac{1}{|1-\Pi|^2} \simeq 1 + 2\,\Delta\alpha(M^2), \tag{6}$$

where $3\alpha/4\pi = 0.00174$ is real emission at all photon energies plus the vertex loop (the large logarithms cancel between them), and Π is the vacuum polarisation of the photon propagator, renormalised at $q^2 = 0$ (α in $\Gamma_{\rm Born}$ is $\alpha(0)$). Quark-side corrections (QCD, $1 - 16\alpha_s/3\pi$) belong to the theory that $\Gamma_{\rm Born}$ is compared with, not to Eq. (6); the J/ψ pole itself is, by definition, not part of Π.

What $\delta(E)$ of `delta_vm` contains. At the full phase space the Ehlotzky–Mitter correction gives $\delta(E) = 0.026135$. This is *exactly* (to 10^{-9}, checked in `validate_rc.py`, section 2) the inclusive QED correction plus the *electron* loop only:

$$\delta(E) = \frac{3\alpha}{4\pi} + 2\,\Delta\alpha_e(M^2) = 0.001742 + 0.024394.$$

The term $(4\alpha/3\pi)\ln(M/m_e)$ that it carries is this loop, not an uncancelled final-state logarithm. The muon, τ and hadronic loops are not in it.

Lepton loops at $s = M^2$. One loop, exact in the lepton mass, time-like ($\Delta\alpha_\ell = \frac{2\alpha}{\pi}\int_0^1 x(1-x)\ln|1 - x(1-x)s/m_\ell^2|\,dx$):

loop	$s/4m^2$	$\Delta\alpha(M^2)$	$\frac{\alpha}{3\pi}\left(\ln\frac{s}{m^2} - \frac{5}{3}\right)$	$1 + 2\Delta\alpha$
e	9.2×10^6	0.012197	0.012197	1.02439
μ	214.8	0.003935	0.003940	1.00787
τ	0.76 (below threshold)	−0.000781	—	0.99844
hadrons	0.0087 (estimate)			1.017

Electron versus muon loop. Both are far above threshold, so both follow the asymptotic form $(\alpha/3\pi)[\ln(s/m^2) - 5/3]$; for the muon the mass correction is only $-0.1\,\%$ of the loop. The electron loop

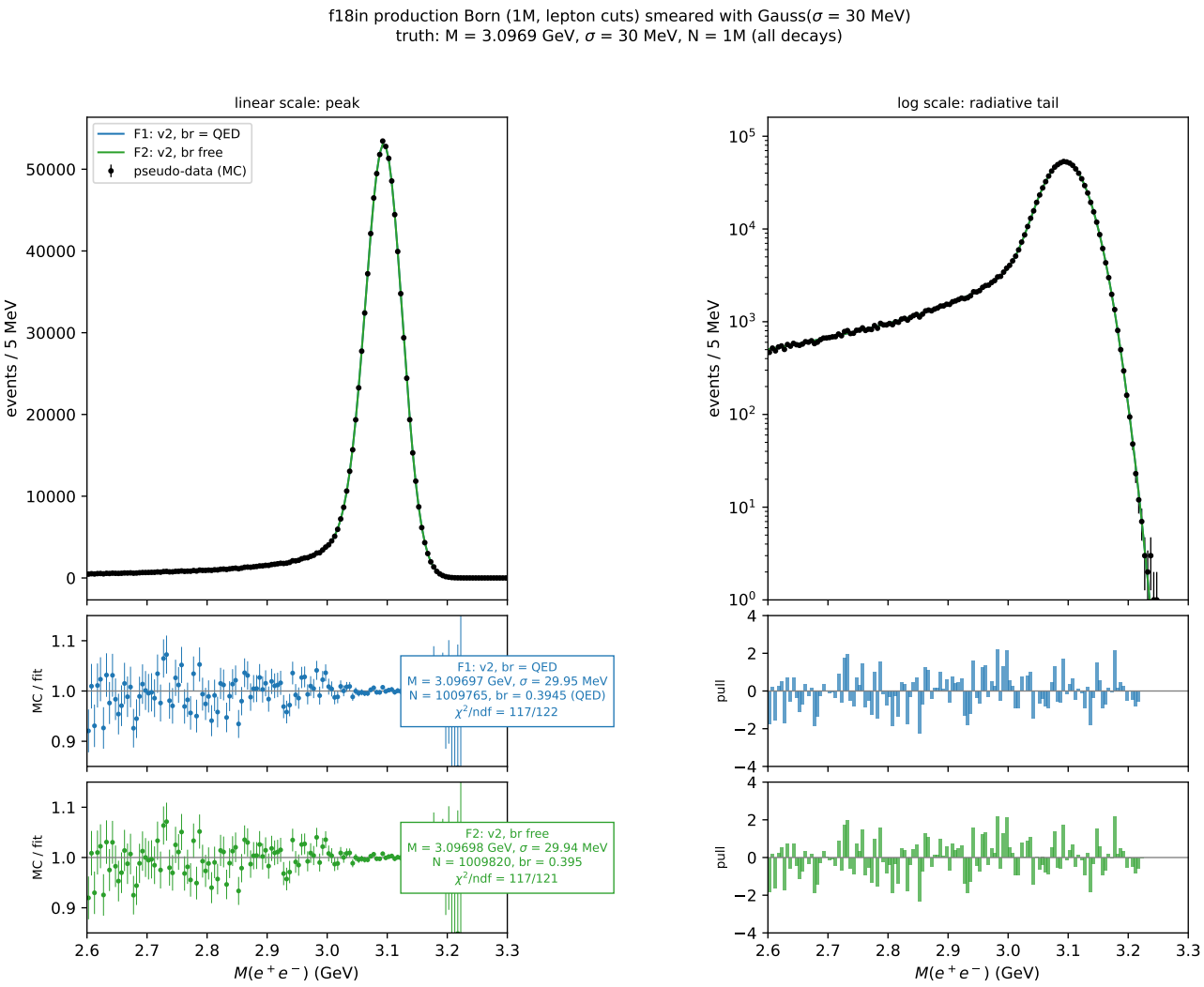


Figure 7: F1 and F2 on the 1M f18in production Born sample (generator lepton cuts), smeared with 30 MeV.

is 3.10 times the muon loop. Their difference does not depend on s:

$$\Delta\alpha_e - \Delta\alpha_\mu = \frac{\alpha}{3\pi}\ln\frac{m_\mu^2}{m_e^2} = 0.008256 \quad \text{(exact one loop: 0.008262)},$$

i.e. 1.65 % in the rate. Leaving the muon loop out of Γ_{Born} is a 0.8 % error, larger than the 0.5 % uncertainty of the PDG branching ratio. The τ loop is below threshold at $M_{J/\psi}$ and small and negative. The hadronic value is the space-like parametrisation of `exclurad_py/core/virtual.py` evaluated at $Q^2 = M^2$, an estimate of the non-resonant part only; a proper number needs a dispersive evaluation of $\Delta\alpha_{\text{had}}$ at $\sqrt{s} = 3.1$ GeV with the J/ψ excluded. Fig. 8 shows the loops against $\sqrt{s}$.

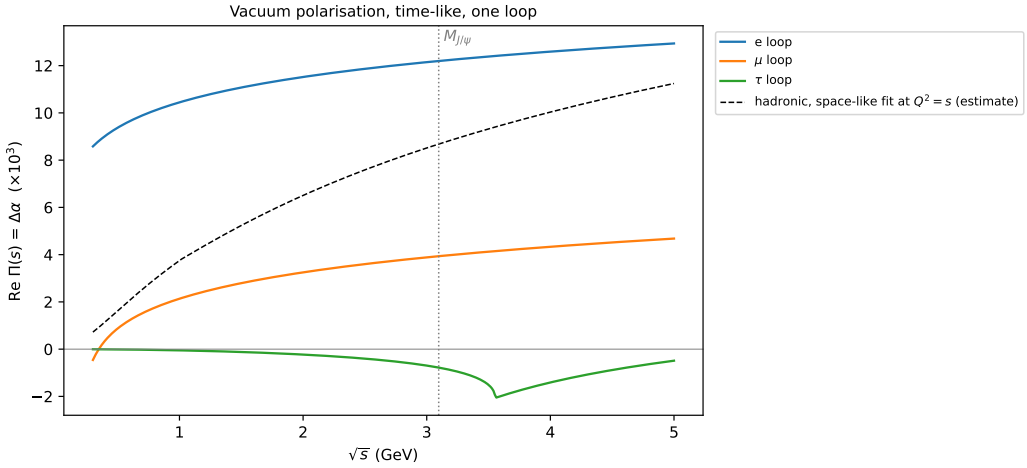


Figure 8: Real part of the one-loop vacuum polarisation for e, μ, τ loops (time-like), and the space-like hadronic fit at $Q^2 = s$ as an estimate.

From Γ to Γ_{Born}.

corrections divided out	$\Gamma/\Gamma_{\text{Born}}$	$\mathcal{B}_{\text{Born}}$ (%)
$3\alpha/4\pi$ only	1.0017	5.961
+ electron loop ($= 1 + \delta(E)$, `delta_vm`)	1.0266	5.816
+ μ, τ loops	1.0332	5.779
+ hadronic (estimate)	1.052	5.68

(computed with $|1 - \Pi|^{-2}$; `mee_lineshape/vp/vp_table.py`). Vacuum polarisation, not radiation, dominates: 5 % against 0.17 %.

Consequences for the generator and the fit. None for normalisation: vacuum polarisation is a common factor of every decay and the generated decays are weighted with the PDG branching ratio, without $1+\delta$. The only trace is in Eq. (4), where the electron-loop constant appears in both numerator and denominator and cancels up to second order: removing it changes F by +0.14 %, i.e. $1 - F(1\text{ MeV})$ from 0.3945 to 0.3937 and $P(M_{ee} < 3.0)$ from 0.1466 to 0.1455. This is far below anything a fit to data can see and is left as it is.

6 Recommendation for fitting data

1. **Do not fit the radiative part.** Use `dNdMee_v2` with `br` < 0: the fraction and the shape of the radiative tail are QED, fixed by M and m_e, and validated against the generator to five digits.

Freeing `br` adds nothing (it returns the QED value) except a correlation with the resolution. This is the "fix the MC parameters, then fit only the resolution" idea, with the MC parameters replaced by their exact values.

2. **Free parameters on data:** the yield N, the mass M (in practice the momentum scale), the resolution σ, and the background.

3. ε**:** keep 1 MeV (or anywhere in 0.1–1 MeV). Moving it does not change the fit as long as $\varepsilon \ll \sigma$ and `br` follows it, which v2 does automatically; above $\sim$5 MeV M starts to be pulled.

4. **The yield** N is the total number of $J/\psi \to e^+e^-(\gamma)$ decays over all M_{ee}; divide it by the PDG branching ratio with no extra $1+\delta$.

5. **Do not use the original `dNdMee.cpp` for yields**: its 2.5 % loss is invisible in χ^2.

What a Gaussian does not cover. The real detector adds its own low-mass tail (bremsstrahlung of the leptons in the target and in the material before the drift chambers), shifts the momentum scale (the 200-event GEMC test of 8 October showed lepton momenta 1.4 % low), and is not Gaussian in the tails. Fitting data with a Gaussian σ alone would push that detector tail into N and M. The next step is therefore GEMC: take the reconstructed-minus-generated M_{ee} of each event from the f18in production, parametrise the response (Gaussian core plus a low-side tail), and fit data with

$$\frac{dN}{dM_{ee}} = N\left[\text{QED (fixed)}\otimes\text{detector response (shape from GEMC, scale and width free)}\right]+\text{background.}$$

For the yield the most robust alternative is a template fit with the GEMC reconstructed M_{ee} of the Born and radiative samples directly.

Files

Branch `mee-lineshape` of `jpsi-rc-2026`, directory `mee_lineshape/`:

- `dNdMee_v2.cpp` — the function;
- `study_lineshape.py` — all fits, through the compiled library;
- `make_figures_mee.py`, `make_truth_nores.py`, `vp/vp_table.py` and this report.

Inputs: 200k Born generator events (exclurad_py, card `born_jpsi.txt`, model `jpsi.diffrad_vm_pdet`, `iacc none`) in `/Volumes/wd_14tb/jpsi_electro_rc/mee200k_0956`, and the 1M Born sample in `/volatile/clas12/vpk/prod_jpsi_f18in_20261009`.